

Compute Services

Use Case 1: Autoscaling and Load Balancing

Scenario: Create a scalable e-commerce website.

Q] Set up an EC2 instance for hosting a simple website.

STEP 1:- Create IAM user

Launch EC2 instance for website host

(t2.micro, t3.micro, t3.nano, m5.large, c5.xlarge, r5.large)

Create security group

(SSH, HTTPS, HTTP)

Q] Configure an Auto Scaling group to handle traffic spikes.

STEP 2:- In EC2 services select Auto Scaling Groups

Create Auto Scaling Group

Click on Launch Template

Create Launch Template

In Auto Scaling Group , select Launch Template

Click to a new Load Balancer

Create target group

Click on Create Auto Scaling Group

Q] Use an Elastic Load Balancer to distribute traffic across instances.

STEP 3:- In EC2 services, select Load Balancer

Compare and Select load balancer type

Select Application Load Balancer

Click on Create

Create Application Load Balancer

Q] Monitor scaling events using CloudWatch.

STEP 4 :- CloudWatch

Search for CloudWatch in search bar

Click on Alarms

Click on Create Alarms

Configure Alarm

Click on Create Alarm

USE CASE 2: Serverless Architecture

Scenario: Build a serverless photo-upload application.

Q] Use S3 for storage.

STEP 1:- Create IAM User

Navigate to S3 Service

Click on Create Bucket

Click on Upload

Q] Trigger Lambda functions for image resizing upon upload.

STEP 2:- Navigate to AWS Lambda Service

Create a Lambda Function

Q] Store metadata in DynamoDB

STEP 3:- Navigate for DynamoDB Table

Create New Table

Click on Create Table

Set some configuration

Click Create to finalize

Q] Use API Gateway to expose the functionality.

STEP 4:- Navigate to Lambda

Create new function

Create Function

Navigate to API Gateway

Create new API

Select HTTP API and Build

Configure API

Click on create API

Setup API

In Action

Select Deploy API

Storage Services

USE CASE 3: Backup and Restore

Scenario: Implement a backup strategy for critical data.

Q] Use AWS Backup to back up RDS and S3 data.

STEP 1:- Navigate to AWS Backup Service

Click on Create Backup Plan

Enable AWS Backup for S3

Create a Backup Plan for S3

Click on Create Plan

Q] Restore data to a new region as part of disaster recovery testing.

STEP 2:- Navigate to AWS Elastic Disaster Recovery

Create Recovery Environment

Launch Recovery Instance

Q] Analyze costs associated with the backup.

STEP 3:- Understand AWS Backup Pricing Structure

Calculate your estimated monthly cost

Monitor Cost

USE CASE 4: Lifecycle Management

Scenario: Optimize storage costs.

Q] Implement S3 lifecycle policies for transitioning objects to Glacier.

STEP 1:- Navigate to S3 service

Select S3 Bucket

Create Lifecycle Rule

Configure settings

Review

Create Rule

STEP 2:- In S3 Bucket

Create it in different region

Create Replication Rule in Source Bucket

Set IAM Roles

Review

Create Rule

NETWORKING

USE CASE 5: Secure Network Architecture

Scenario: Design a VPC for a multi-tier web application.

Step 1: Create a VPC

Step 2: Create Subnets

- Create Public Subnet

- Create Private Subnet

Step 3: Configure the Internet Gateway

- Create an Internet Gateway

- Attach the Internet Gateway to the VPC

Step 4: Configure Route Tables

- Public Subnet Route Table

- Associate the route table with the Public Subnet.

- Private Subnet Route Table

- Associate this route table with the Private Subnet.

Step 5: Configure a NAT Gateway

Create an Elastic IP

Create a NAT Gateway

Update Private Route Table

DATA AND ANALYTICS

USE CASE 6: Real-Time Data Processing

Scenario: Analyze streaming data from IoT devices.

Q] Use Kinesis Data Streams to ingest data.

Step 1: Set Up Kinesis Data Streams

Open Kinesis Console:

- Go to Kinesis Dashboard > Data Streams.

Create a Stream

Click Create Data Stream

Step 2: Set Up IoT Device Data Ingestion

Q] Use Lambda or Glue for data processing.

Step 3: Set Up Data Processing

Using AWS Lambda

- Open Lambda Console
- Create a Lambda Function
- Configure Kinesis as the Trigger
- Write Processing Code

Step 4: Store Processed Data in S3

Create an S3 Bucket

Write Data to S3

Q] Store processed data in S3 and visualize it using QuickSight.

Step 5: Visualize Data Using QuickSight

Open QuickSight Console

Configure QuickSight

Prepare the Data

Create Visualizations

Step 6: Optimize and Monitor

Enable Monitoring

- Use CloudWatch to monitor Kinesis, Lambda, and Glue performance.

Optimize Costs

- Use lifecycle policies to archive old data in S3 (e.g., move to S3 Glacier).

USE CASE 7: Data Lake Implementation

Scenario: Set up a centralized data repository.

Q] Use S3 as the data lake.

Step 1: Set Up an S3 Bucket as the Data Lake

Open the S3 Console

Navigate to the S3 Dashboard

Create a Bucket

Enable bucket versioning for data protection.

Click Create Bucket.

Organize Data

Create folders in the bucket for different datasets

Upload sample data files to the appropriate folders.

Q] Catalog the data using Glue.

Step 2: Catalog the Data Using AWS Glue

Navigate to the AWS Glue Dashboard

Create a Crawler

Verify the Catalog

Q] Query data with Athena.

Step 3: Query Data with Amazon Athena

Navigate to the Athena Dashboard.

Set Up Query Results Location

Run Queries

Q] Secure access with IAM and bucket policies.

Step 4: Secure the Data Lake

Create IAM Role for Glue

Create IAM Role for Athena

S3 Bucket Policies

Enable S3 default encryption (e.g., SSE-S3 or SSE-KMS) to encrypt all data in the bucket.

Step 5: Test and Validate

Test Data Access

Verify Security

Step 6: Monitor and Optimize

Enable Logging:

- Enable S3 server access logging or CloudTrail to monitor access to the data lake.

Optimize Costs:

- Use S3 Lifecycle Policies to transition older data to S3 Glacier.

Track Usage:

- Use CloudWatch or AWS Cost Explorer to monitor Glue and Athena usage.

MACHINE LEARNING

USE CASE 8: Personalized Recommendation

Scenario: Build a recommendation engine.

Q] Use Amazon Personalize to create a model.

STEP 1: Data Preparation and Loading

1. Gather your data

Users; Items; Interactions

2. Upload Data to S3

3. Create a Dataset Group

4. Import Datasets

STEP 2: Create a Solution

1. Select a Recipe

After your datasets are imported successfully, click on create solution

2. Configure Solution Setting

3. Create Solution Version

Once the solution is created, click on it and select create solution version

Q] Train the model using sample data.

STEP 3: Create Campaign

1. Launch Campaign

After creating a solution version, navigate back to your solution and click on create campaigns

2. Configure Campaign Setting

Q] Deploy the model and integrate recommendations into an application.

STEP 4: Integrate Recommendations into your Application

1. Get Recommendations via API

2. Sample Code for API call

3. Test your Recommendations

USE CASE 9: Continuous Deployment

Scenario: Set up a CI/CD pipeline.

Q] Use CodePipeline, CodeBuild, and CodeDeploy.

Q] Deploy updates to an application hosted on EC2.

STEP 1 : EC2 instance

IAM Role :- for CodePipeline , CodeBuild , CodeDeploy

STEP 2 : S3 Bucket

STEP 3 : Create CodeDeploy Application

Create Deployment Group

Create a CodeBuild Project

STEP 4 : In the AWS CodeBuild console

Create Project

STEP 5 : Create Pipeline

In CodePipeline console
service role

Q] Rollback automatically on deployment failures.

STEP 6 : For Rollback

While creating a Deployment Group select a automatic rollback is enabled.

Select Rollback on failure

Use Case 10: Infrastructure as Code

Scenario: Automate infrastructure provisioning.

Q] Use CloudFormation to create a stack for a web application.

STEP 1 : Navigate to AWS CloudFormation

Create Stack

Upload a template file

Stack details

Create Stack

After Stack Created

Q] Include EC2, RDS, and an S3 bucket in the stack.

STEP 2 : Navigate to Outputs

Check for EC2, RDS, S3

Q] Update the stack to add a new feature.

STEP 3 : Go to CloudFormation Console

Select Stack you want to update

Click Update

Choose Replace current template

Click on Next

Click Update Stack